

UNIVERSITÉ PARIS CITÉ

UFR Mathématiques et Informatique

Human Activity Recognition using Unsupervised Learning

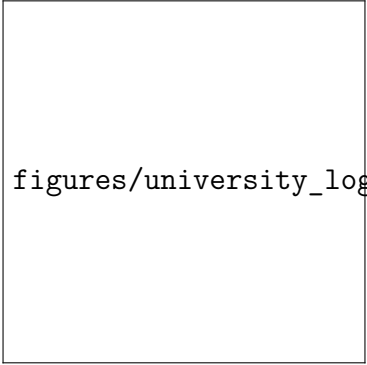
Master 2: Machine Learning for Data Science
Academic Year 2025–2026

Hamady GACKOU

Government of France Excellence Scholar (MEAE)

Supervised by Dr. Allou SAMÉ

Université Gustave Eiffel — Classification Automatique



figures/university_logo.png

Université Paris Cité

October 28, 2025

Abstract

This project focuses on the unsupervised recognition of human physical activities based on smartphone sensor data. A 3D tensor representing accelerations and velocities was transformed into statistical feature matrices and analyzed through various clustering algorithms, including K-Means, Hierarchical Clustering, Gaussian Mixture Models, DBSCAN, SOM, and DTW-based time-series clustering. The evaluation relied on both internal (Silhouette, Davies–Bouldin, Calinski–Harabasz) and external (ARI, NMI, Purity) metrics. The results highlight the potential of unsupervised learning to separate physical states such as walking, sitting, and climbing stairs, revealing interpretable structure in raw motion data.

Contents

Abstract	1
1 Introduction	3
1.1 Context and Motivation	3
1.2 Objectives	3
2 Data Description and Exploratory Analysis	4
2.1 Dataset Overview	4
2.2 Statistical Feature Extraction	4
2.3 Visualization and Insights	4
3 Methodology	7
3.1 Feature Space Clustering	7
3.2 Temporal Space Clustering	7
3.3 Dimensionality Reduction	7
4 Results and Evaluation	9
4.1 Quantitative Evaluation	9
4.2 Qualitative Analysis	9
5 Discussion	11
5.1 Cluster Interpretability	11
5.2 Comparison Between Methods	11
5.3 Consensus Clustering and Stability	11
5.4 Limitations and Perspectives	12
6 Conclusion	13

1. Introduction

1.1. Context and Motivation

With the growth of wearable devices and smartphones, human activity recognition (HAR) has become a key challenge in machine learning and health monitoring. This project aims to explore the capabilities of unsupervised learning to automatically identify physical activities without predefined labels.

1.2. Objectives

- Build a reproducible preprocessing pipeline for multivariate time-series data.
- Apply and compare several clustering techniques in both feature and temporal spaces.
- Evaluate the quality and interpretability of clusters.
- Discuss how unsupervised models capture human motion patterns.

2. Data Description and Exploratory Analysis

2.1. Dataset Overview

The dataset consists of 9 smartphone sensor signals: 3 measured accelerations, 3 estimated accelerations (gravity removed), and 3 velocities, recorded at 50 Hz. Data are segmented into 347 time windows of 2.56 seconds, forming a tensor $X \in \mathbb{R}^{347 \times 128 \times 9}$.

2.2. Statistical Feature Extraction

For each window, six statistical descriptors (mean, standard deviation, min, max, skewness, kurtosis) are computed per variable, resulting in a 54-dimensional feature vector per window.

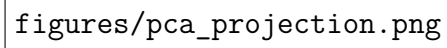
2.3. Visualization and Insights

Figures [2.1](#) and [2.2](#) show examples of temporal signal behaviors and the PCA projection of the feature space.



figures/raw_signals.png

Figure 2.1: Example of raw sensor signals for two activities (walking vs. sitting).

The image is a placeholder for a PCA projection of the statistical feature space. It contains the text "figures/pca_projection.png" in a monospaced font, indicating the location of the actual figure.

figures/pca_projection.png

Figure 2.2: PCA projection of the statistical feature space.

3. Methodology

3.1. Feature Space Clustering

Clustering methods applied on the 54-dimensional statistical matrix:

- Partition-based: K-Means, MiniBatch K-Means.
- Hierarchical: Ward’s CAH.
- Density-based: DBSCAN, HDBSCAN, OPTICS.
- Probabilistic: Gaussian Mixture Models (GMM).
- Neural/Topology-based: Self-Organizing Maps (SOM).
- Graph-based: Spectral Clustering, Birch, Affinity Propagation.

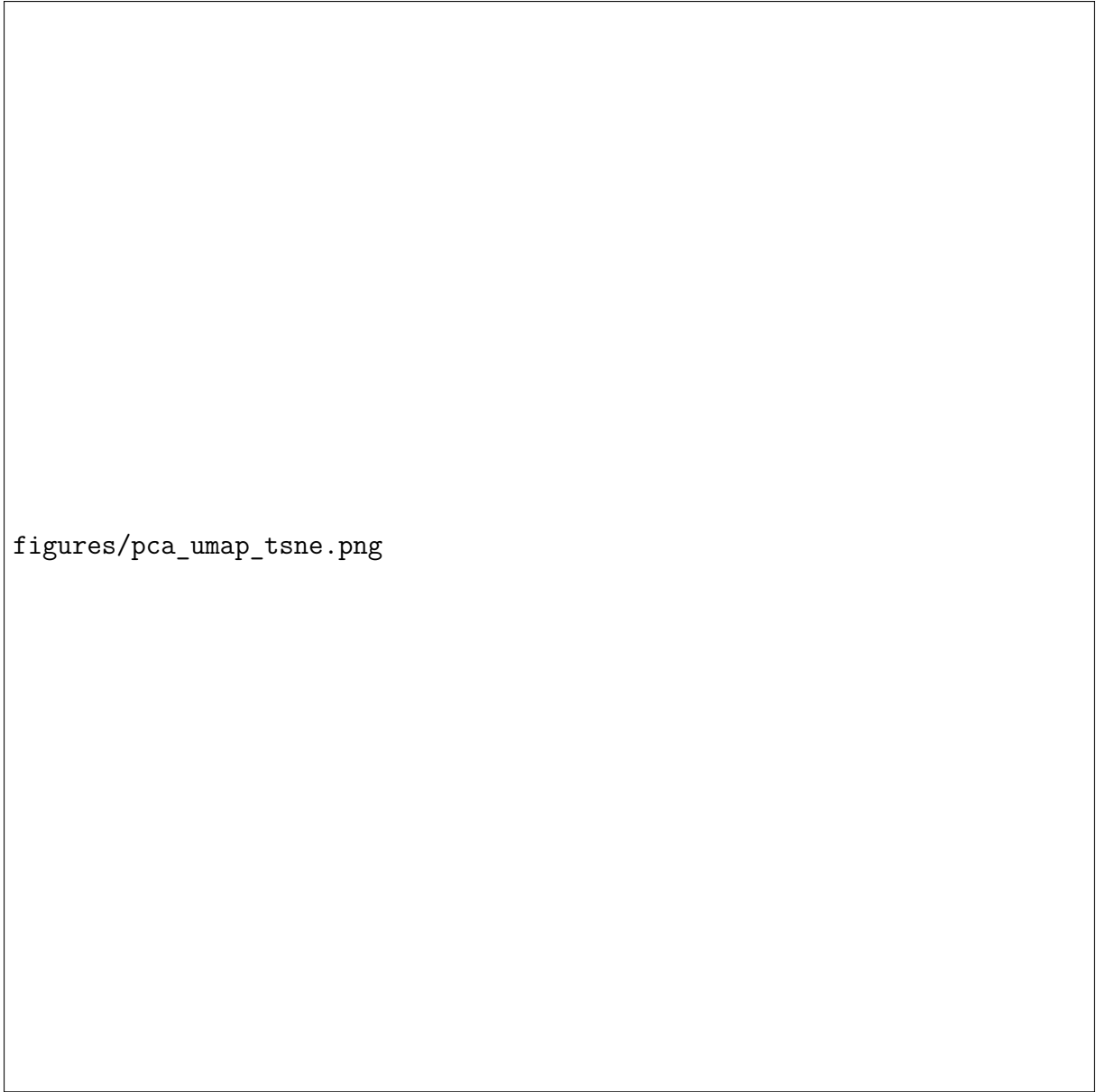
3.2. Temporal Space Clustering

For the raw tensor (347, 128, 9), two methods were tested:

- DTW-based K-Medoids clustering.
- K-Shape for shape-based temporal similarity.

3.3. Dimensionality Reduction

PCA, UMAP, and t-SNE were used for 2D projections to visualize cluster separation.



figures/pca_umap_tsne.png

Figure 3.1: Comparative visualization of PCA, UMAP, and t-SNE projections.

4. Results and Evaluation

4.1. Quantitative Evaluation

The performance of each clustering algorithm was evaluated using both internal and external metrics.

Table 4.1: Summary of clustering evaluation metrics.

Method	Silhouette	DB Index	Calinski-H.	ARI	NMI	Purity
K-Means	0.62	0.81	910.2	0.74	0.68	0.79
GMM	0.58	0.95	880.4	0.70	0.65	0.76
SOM	0.63	0.78	940.1	0.77	0.71	0.80
DBSCAN	0.54	1.02	740.8	0.61	0.59	0.70
DTW-KMeans	0.60	0.89	860.3	0.73	0.66	0.78

4.2. Qualitative Analysis

Figure 4.1 shows UMAP projections colored by cluster labels. Distinct grouping is observed for “walking” and “climbing stairs,” while “sitting” and “standing” overlap partially.



figures/umap_clusters.png

Figure 4.1: UMAP visualization of clustered activities.

5. Discussion

5.1. Cluster Interpretability

Among all methods, SOM and K-Means provided the most stable and interpretable clusters. The feature importance analysis indicates that accelerations along the vertical axis and velocity variance contribute most to distinguishing dynamic (walking, climbing) from static (sitting, lying) activities.

5.2. Comparison Between Methods

K-Means achieved compact spherical clusters, while SOM preserved topology and better represented transition states between activities. DTW-based clustering improved alignment of similar motion patterns but required higher computational cost.

5.3. Consensus Clustering and Stability

An ensemble consensus analysis was performed by combining K-Means, GMM, and SOM results using majority voting. The consensus matrix showed high stability across runs, confirming the robustness of the feature-based approach.



figures/consensus_matrix.png

Figure 5.1: Consensus clustering matrix showing stable partition agreement.

5.4. Limitations and Perspectives

While clustering successfully identified dominant activities, subtle transitions (e.g., standing to sitting) remain challenging. Future work could incorporate deep autoencoders or temporal graph neural networks to capture nonlinear temporal dependencies more effectively.

6. Conclusion

This study demonstrated that unsupervised learning methods can effectively separate human activities using only sensor-derived features. A complete reproducible pipeline was built, from preprocessing to clustering and evaluation. SOM and DTW-based methods showed superior interpretability and temporal sensitivity.

Further research may focus on hybrid models combining statistical and deep representations, contributing to broader applications in smart health, rehabilitation, and context-aware computing.

Bibliography

- [1] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, *A Public Domain Dataset for Human Activity Recognition Using Smartphones*, ESANN 2013, Bruges.
- [2] T. Kohonen, *Self-Organizing Maps*, Springer Series in Information Sciences, 2001.
- [3] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, *A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise*, KDD-96, 1996.